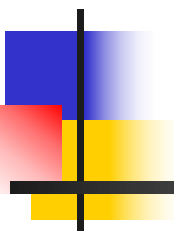


Lecture 1

History and scope of Microbiology

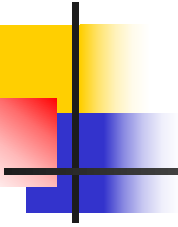




Objectives for this lecture:

- At the end of this lecture, you should be able to:
 - Define a microorganism
 - Define the spontaneous generation theory and explain how this idea was disproved
 - Explain the development and significance of microscopy in discovering microorganisms
 - Explain Koch's postulates
 - Describe the interaction of plants with microbes giving examples
 - Know the names of the main early microbiologists and be able to briefly describe their major contributions including century
 - **Van Leeuwenhoek; Louis Pasteur; Semmelweiss (山姆维斯); Robert Koch; Beijerinck (拜耶林克); Winogradsky (维诺格拉斯基); Alexander Fleming**

Main reference: Prescott's Microbiology. Chapter 1.



Microbiology

- Definition: The study of microorganisms (microbes)
 - defined as organisms too small to be seen by the naked eye (but there are exceptions e.g. multicellular microbes)
 - includes bacteria, archaea (古細菌), fungi (molds and yeasts), viruses, algae, and protozoa (原生動物)
 - involves understanding:
 - Where are they found?
 - What is their structure?
 - Why do some cause disease while others are beneficial?
 - What is their role in the environment?
 - How can we make use of them?

- Aristotle (384-322) and other philosophers of Ancient Greece
 - **Spontaneous generation:** living organisms could develop from non-living materials.
- Illnesses believed to be caused by
 - Supernatural forces
 - Miasmas (vapours;瘴气)
 - Imbalance of the four humours (blood, phlegm痰, choler胆汁, melancholy愁思)
- Complete inability to control infectious diseases
 - Black death (plague): 25 million died 1347-1352 Europe
 - Smallpox (viral) and cholera (霍乱 bacterial) and numerous other diseases killed millions



"The Doctor Beak of Rome" engraving by Paul Fürst.
The beak is a primitive gas mask, stuffed with substances (such as spices and herbs) thought to ward off the plague.

Pre-Microbiology era (pre 1660)

- Animal husbandry (饲养)
 - Sudden death in animals
- Agriculture
 - Crop failure
- Food production
 - Fermented foods
 - Cheese, yogurt, beer, wine
- Spoilage
 - All fresh foods eaten immediately
 - Short term storage for dried and pickled goods



Farmers and livestock shared living space



Trade in spices drove the economies of the world. Spices were essential for pickling and salting food for storage.

Development of microscopy

1590: Hans and Zacharias Janssen

- mounted two lenses in a tube to produce the first compound microscope (3-10 fold magnification)

1660: Robert Hooke (1635-1703)

- published first drawings of microorganisms

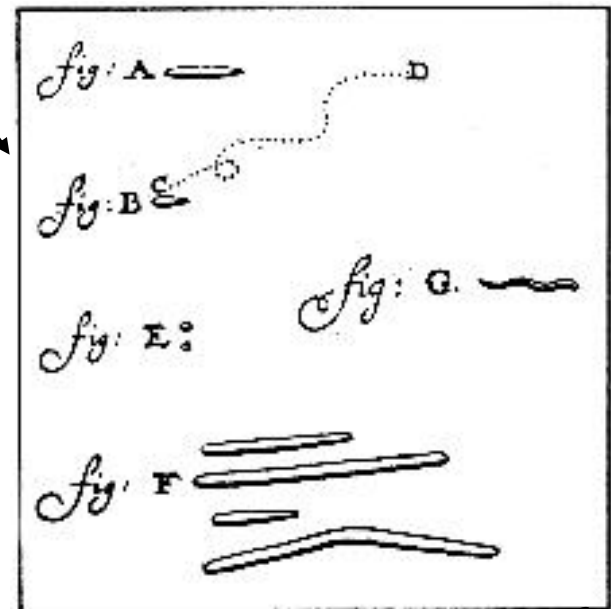
1676: Anton van Leeuwenhoek (1632-1723)

- amateur microscopist (draper [布商] in Netherlands)
- first person to observe microorganisms (70 -300 fold magnification)

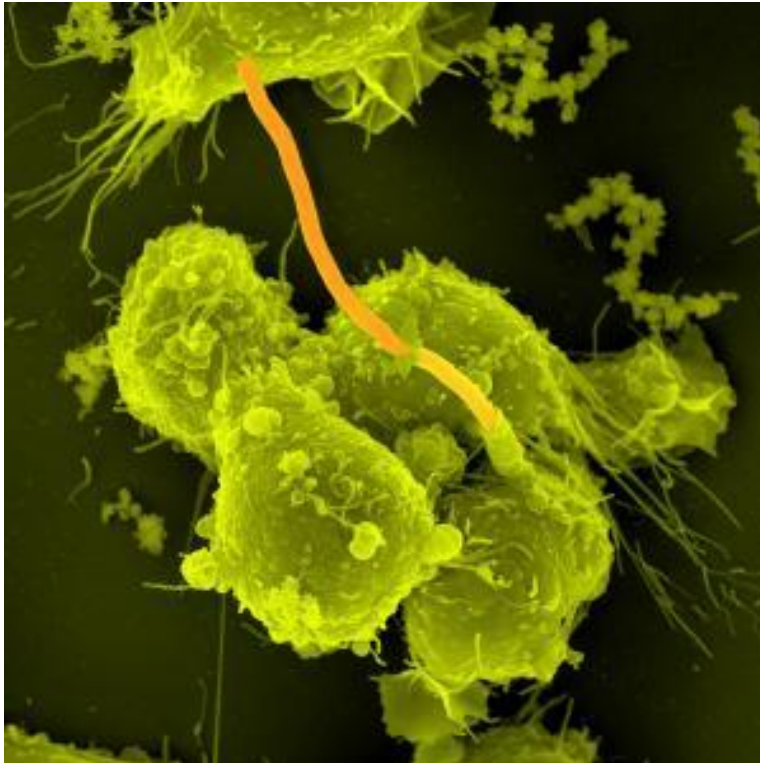
1883: Carl Zeiss and others

- improved microscopes (such as immersion lenses [油浸物镜] and lenses which reduce chromatic aberration [色差])

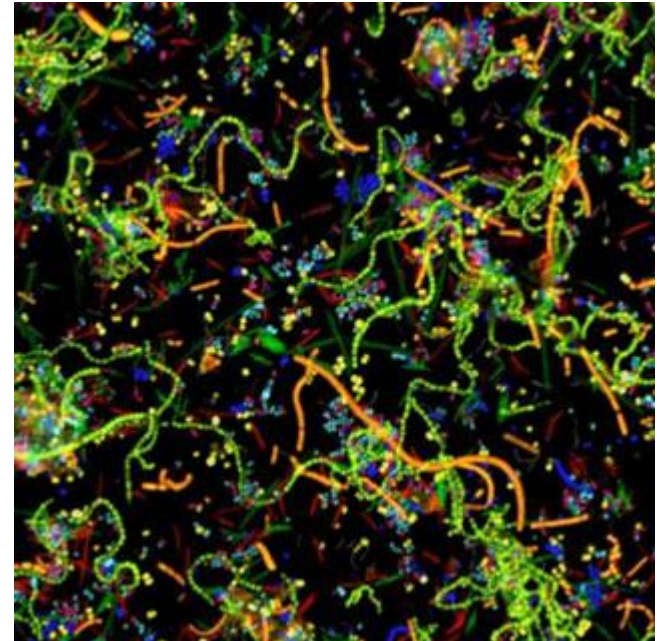
1931: the first electron microscope (100,000 fold magnification)



Modern images of microorganisms



Bacteria (orange) invading a human cell (green).
(scanning electron micrograph, falsely coloured)



A mixture of microbes from oral cavity.
(fluorescence microscopy with probes labelled a different colour for each microbial species)

<http://www.biotechniques.com/news/CLASI-FISHing-for-microbes/biotechniques-315946.html>

Spontaneous generation controversy

1688: Francesco Redi (1626-1697)

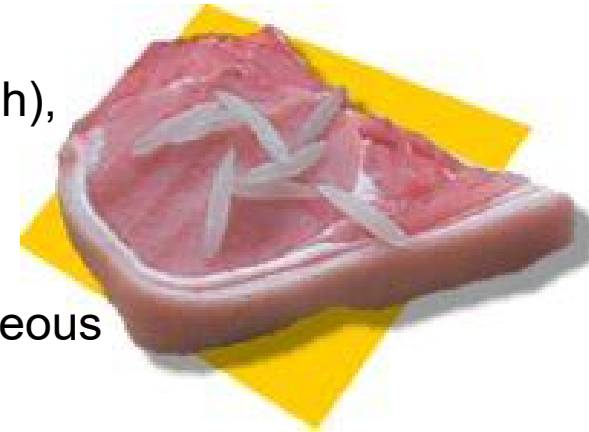
- refuted the idea of **spontaneous generation** by showing that rotting meat carefully kept from flies will not spontaneously produce maggots (蛆)

1748: John Needham

- If flasks of meat broth were boiled long enough (1-2 h), and then closed, microorganisms did not grow

1861: Louis Pasteur (1822-1895)

- proved that microorganisms do not arise by spontaneous generation



Concepts

- Sterilization: a method that kills all living organisms (灭菌)
- Aseptic technique: a series of practices to reduce contamination of a sterile site by microbes (无菌技术)

Pasteur's experiment with Swan-neck flasks

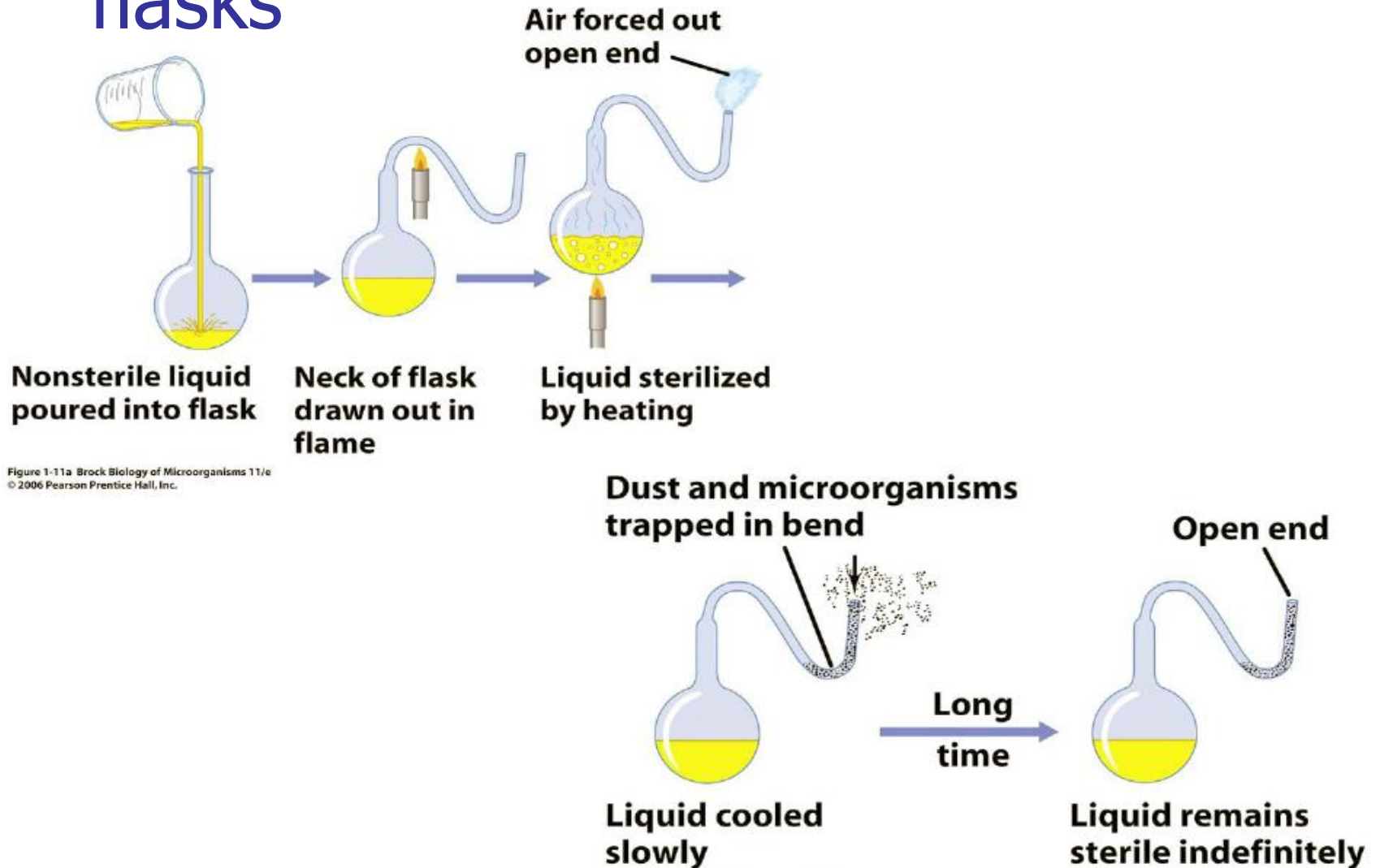


Figure 1-11a Brock Biology of Microorganisms 11/e
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Proof that microbes cause disease

1835: Agostino Bassi de Lodi

- showed that a disease affecting silkworms was caused by a fungus - the first microorganism to be recognized as a contagious (感染性的) agent of animal disease

1847: Ignaz Semmelweis (1818-1865)

- demonstrated that simply washing hands between delivering babies, significantly reduced the spread of childbed fever(产褥热)

1857: Louis Pasteur

- proposed the "germ theory" of disease



Germ theory of disease

1876: Robert Koch (1843-1910)

- cultivated anthrax (炭疽) bacteria outside the body using blood serum at body temperature
- first direct demonstration that bacteria cause disease by injecting mice

1877: Julius Richard Petri

- designed a flat circular dish with lid (Petri dish)

1880s: Walter Hesse

- used agar as a solidifying agent
 - melts at 100°C, solid at 37°C

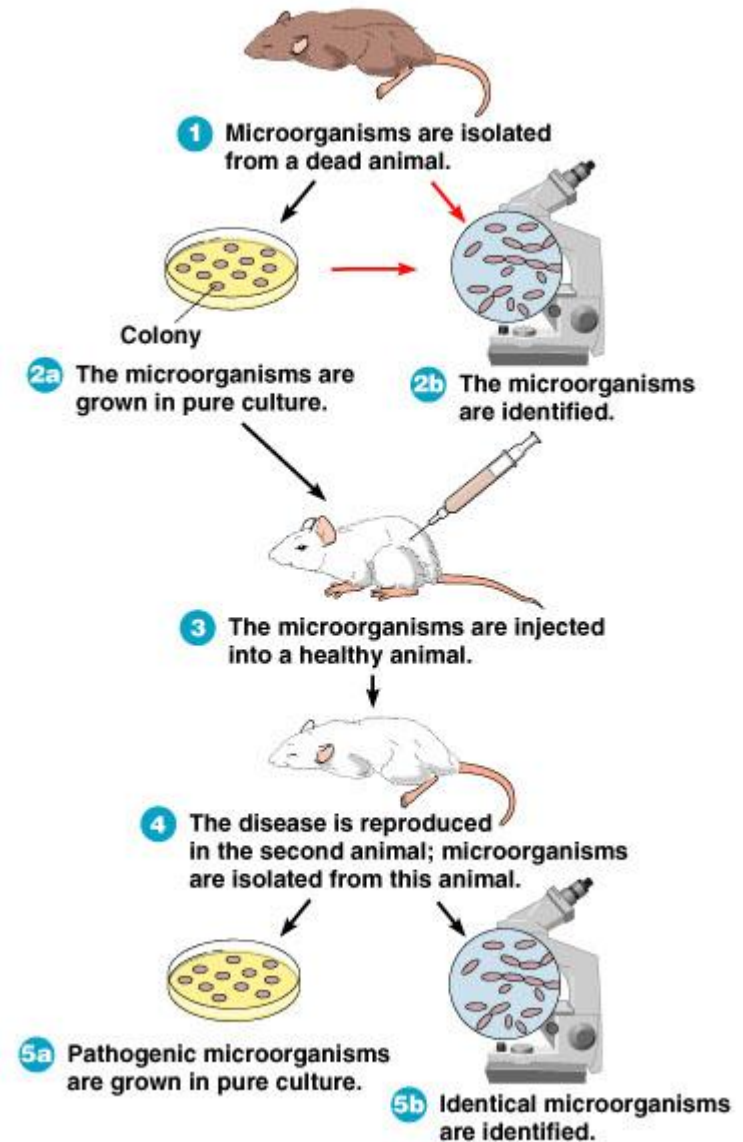
Concept

- Pure culture: population of cells that are identical because they arise from a single cell
 - allows isolation of a single microorganism



Koch's postulates

- Published "**Koch's postulates**" (1884), the critical test to prove the involvement of a microorganism in a disease
 1. The bacteria must be present in every case of the disease.
 2. The bacteria must be isolated from the host with the disease and grown in pure culture.
 3. The specific disease must be reproduced when a pure culture of the bacteria is inoculated into a healthy susceptible host.
 4. The bacteria must be recoverable from the experimentally infected host.
- **BUT**, there are exceptions to every rule
 - e.g. leprosy (麻风) bacillus cannot be isolated in pure culture



Environmental microbiology

Culture-based techniques started to be applied to the study of microbes in soil and water

Martinus Beijerinck (1851-1931)

- developed enrichment culture technique, a way to isolate microbes with certain growth preferences

Concept

- Selective media: microbes can be cultured selectively by providing nutrients required by some but not all microorganisms in a sample

Sergei Winogradsky (1856-1953)

- isolated ***nitrifying bacteria*** in soil that use atmospheric nitrogen for growth

Concept

- Autotrophic growth: using CO₂ as the sole source of carbon



Winogradsky column

Different microbes growing in different layers of the column, depending on nutrient levels and oxygen levels, as in a lake or salt-marsh

Plants and microbes

1845 to 1892

- Potato blight (枯萎) in Ireland (1845-8) due to fungal disease
- Rust and smut (黑穗病) fungi caused disease in cereals

Martinus Beijerinck

- Nitrogen fixation in root nodules performed by the bacterium, *Rhizobium*

Concept

- Symbiosis: The close association of two organisms. Each partner gains from the relationship
- e.g. symbiosis between *Rhizobium* (根瘤菌) and plants such as peas, beans
- published the first evidence of ***tobacco mosaic virus***



Source of blight:
the fungus
*Phytophthora
infestans*



Nitrogen fixing
nodules fix
atmospheric
nitrogen to
ammonia which is
used by the host
plant for growth.

How has Microbiology changed the way we live?

- The modern city
 - 1860: Separation of drinking water and sewage
 - 1866: Introduction of the flushing toilet
- The control of disease
 - Prevention
 - **1798: Edward Jenner** performs the first vaccination against smallpox
 - **1867: Joseph Lister** introduces aseptic techniques to surgery
 - Treatment
 - **1910: Paul Ehrlich** introduces chemotherapy for syphilis (梅毒)
 - **1929: Alexander Fleming** discovers first antibiotic, penicillin



Victorian London



Smallpox was eradicated from the world in 1980

How has Microbiology changed the way we live?

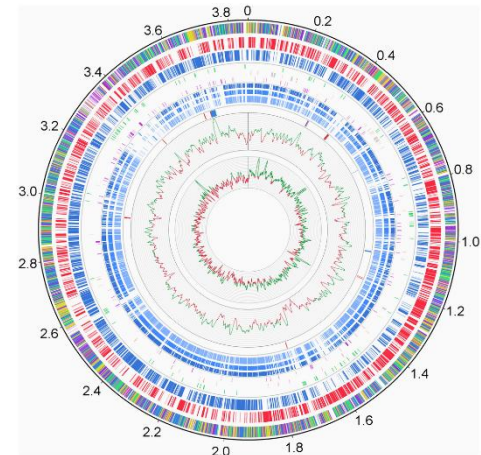
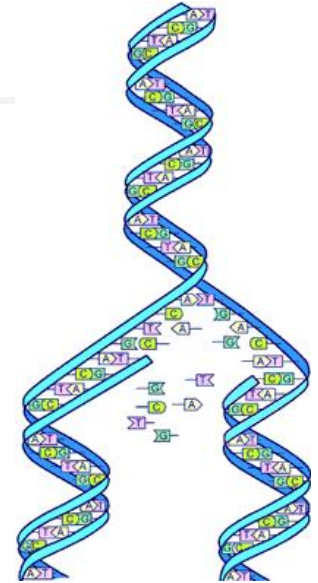
- Prolonged storage of foods
 - 1805: start of mechanical refrigeration
- Industrial scale food production
 - 1900 Dr Ilya (伊利亚) Metchnikoff isolated bacillus from yoghurt
 - 1927 first industrial yoghurt manufacture



Modern Supermarket

Microbiology today

- Molecular and genomic methods for studying microbes are important recent developments
 - 1941: discovery of DNA as the genetic material
 - 1966: discovery of genetic code
 - 1972: gene cloning
 - 1995: whole genome sequencing
- We are in a time of rapid advancement in knowledge that rivals the golden age of microbiology (Koch, Pasteur etc.)
 - a second golden age?



Microbiology: A living study

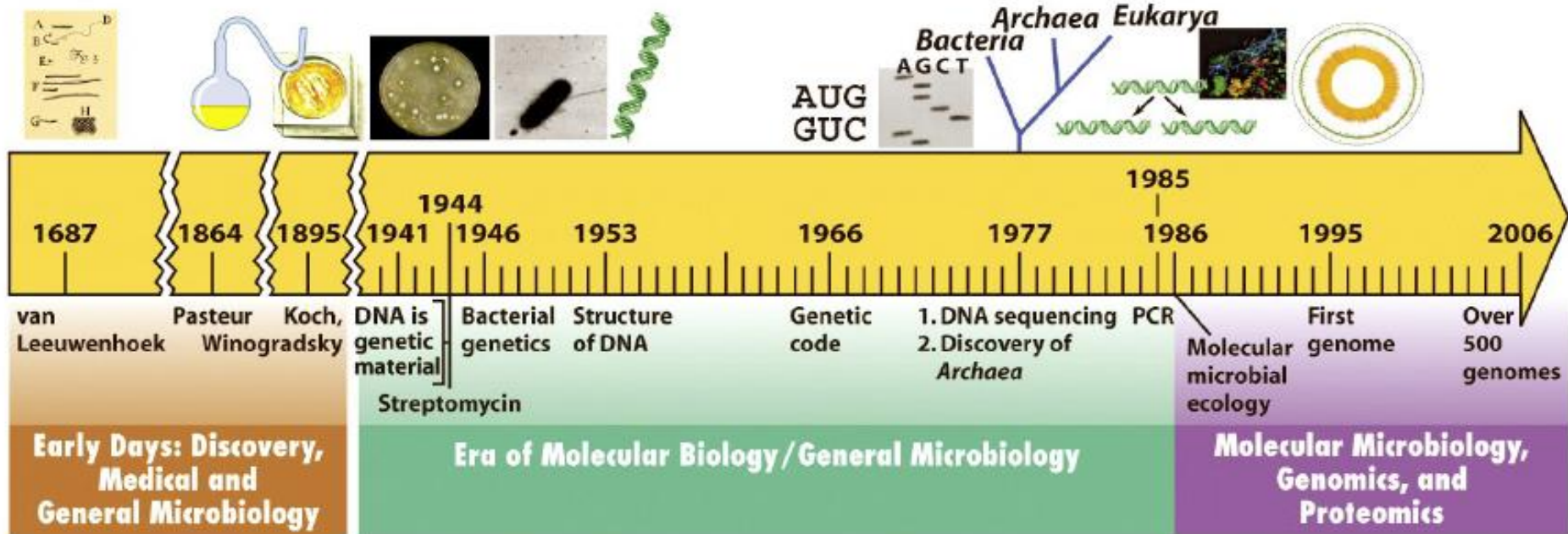
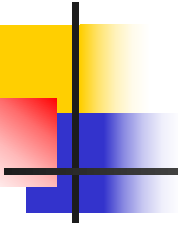
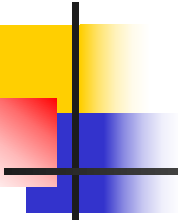


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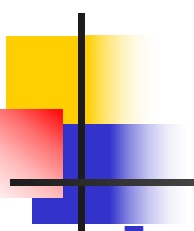
Highlights in the History of Microbiology

- . 1677
 - Observed "little animals" (Antony Leeuwenhoek)
- 1796
 - First scientific Small pox vaccination (Edward Jenner)
- 1850
 - Advocated washing hands to stop the spread of disease (Ignaz Semmelweis)
- 1861
 - Disproved spontaneous generation (Louis Pasteur)
- 1862
 - Supported Germ Theory of Disease (Louis Pasteur)
- 1867
 - Practiced antiseptic surgery (Joseph Lister)



Highlights in the History of Microbiology

- 1876
 - First proof of Germ Theory of Disease with *B. anthracis* discovery (Robert Koch)
- 1881
 - Growth of Bacteria on solid media (Robert Koch)
- 1882
 - Outlined Kochs postulates (Robert Koch)
- 1882
 - Developed acid-fast Stain (Paul Ehrlich)
- 1884
 - Developed Gram Stain (Christian Gram)
- 1885
 - First Rabies vaccination (Louis Pasteur)
- 1887
 - Invented Petri Dish (R.J. Petri)
- 1892
 - Discovered viruses (Dmitri Iosifovich Ivanovski)



Highlights in the History of Microbiology

- 1899
 - Recognized viral dependence on cells for reproduction (Martinus Beijerinck)
- 1900
 - Proved mosquitoes carried the yellow fever agent (Walter Reed)
- 1910
 - Discovered cure for syphilis (Paul Ehrlich)
- 1928
 - Discovered Penicillin (Alexander Fleming)
- 1977
 - Developed a method to sequence DNA (W. Gilbert & F. Sanger)
- 1983
 - Polymerase Chain Reaction invented (Kary Mullis)
- 1995
 - First microbial genomic sequence published (H. influenzae) (TIGR)

Major fields in microbiology

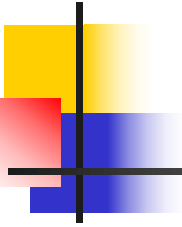
- Medical microbiology and public health
- Industrial microbiology and biotechnology
- Food and dairy microbiology
- Environmental microbiology/microbial ecology
 - Agricultural microbiology
 - Marine microbiology

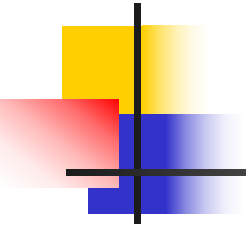




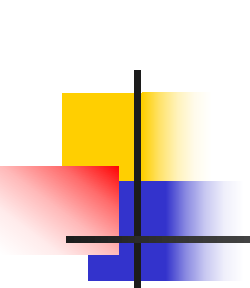
Why study Microbiology?

- Microbes were the first life on Earth
- Microbes created the biosphere that allowed multicellular organisms to evolve
- Multicellular organisms evolved from microbes
- >50% of the biomass on earth is comprised of microbes
- Microbes are everywhere
- Microbes will be on Earth forever

- 
-
- Our understanding of life has arisen largely from studies of microorganisms (biochemistry and genetics)
 - Studies of microorganisms continue to contribute to fundamental knowledge of life processes
 - We still know very little about the microorganisms that are present on Earth



Always trust a microbiologist because they have the best chance of predicting when the world will end.



请同学们举几个例子

- ✓ Health
- ✓ Agriculture
- ✓ Food
- ✓ Environment
- ✓ Industry



